

Qwen3-0.6B Quantization

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Step 1. RTN baseline

- Sweep: granularity (per tensor / channel / g128), sym vs asym
- Granularity dominates: per tensor diverges, per channel 0.26, g128 required
- Asym +0.065 over sym (skewed weight groups)
- Baseline below the 0.36 requirement

W4 g128 asym

26.57%

size (3.76×)

34.53

ppl (FP16 28.65)

0.3392

MMLU (FP16 0.4056)

Step 2-1. RTN + clip

- Shrink each group's grid range: few outliers saturate, the rest get a denser grid
- Range searched per group, errors weighted by Hessian diagonal
- Works as a regularizer on outlier weights; especially effective on a small model
- Notable: MMLU rises above FP16, but the ppl gain is small (still 2.9 above FP16)

Step 1 → RTN + clip

34.53 → 31.53

ppl

0.3392 → 0.4156

MMLU (worst seed 0.4091)

Size unchanged. Full MMLU 0.4176 vs FP16 0.4015.
Kept as MMLU-max alternative.

Step 2-2. GPTQ + AWQ

- Replace plain RTN rounding with GPTQ [1] and AWQ [2], keeping the clip search
- GPTQ: column-wise error compensation; shares the Hessian with the clip search
- AWQ: protect activation-salient channels, folded offline
- Result: both metrics good and balanced. ppl 1.65 better, MMLU still above 0.36
- AutoRound [5], k-means codebook [6], FSQ [9] implemented and measured: none better

RTN+clip → GPTQ+clip+AWQ

31.53 → 29.88

ppl

0.4156 → 0.3694

MMLU

External anchor: official AutoAWQ checkpoint [14],
same protocol: ppl 33.99, MMLU 0.3692.

Step 3. Mixed precision: KL knapsack

- Sensitivity = output KL vs FP, one layer quantized at a time
- Knapsack over 3/4/5/6/8 bit, cost = graded size formula
- v_proj gets the most bits: rank 3 but smallest matrix, cheapest KL per bit

Layer	Params	Mean bits	Rank
q_proj	2.10M	3.79	7
k_proj	1.05M	4.07	6
v_proj	1.05M	4.93	3
o_proj	2.10M	3.79	5
gate_proj	3.15M	3.75	4
up_proj	3.15M	4.18	2
down_proj	3.15M	3.96	1

Step 2 → Step 3

29.88 → 29.72

ppl

0.3694 → 0.3832

MMLU (worst seed 0.3629)

26.50% (3.77×), 4.240 b/w. Submitted model.

Step 4. W4A8

What was done

- Step 2 W4 weights kept; activations entering each linear layer quantized to INT8 with a per-token dynamic scale
- Per-tensor static scale and INT4 activations measured as comparisons

Why activation quantization is hard in LLMs

- Outlier channels: max channel 2642× the median, so an INT8 scale leaves the rest under one level. Weights are near-Gaussian (kurtosis 0.17). LLM.int8 [15], SmoothQuant [3]
- Scale axis: a per-input-channel scale cannot leave the dot product. Only per-tensor or per-token, the axes outliers affect
- Dynamic: no pre-compensation (no GPTQ, no clip search). Static scale +1.7 ppl; dynamic scale costs a reduce per token

W4 → W4A8

29.88 → 29.91

ppl, per-token INT8

0.3694 → 0.3396

MMLU, below 0.36

180

ppl with INT4 activations

Results

Method	Size %	ppl ↓	MMLU 0.1 ↑	MMLU full ↑	Note
FP16	100	28.65	0.4056	0.4015	baseline
Step 1 RTN	26.57	34.53	0.3392	0.3414	
Step 2-1 RTN + clip	26.57	31.53	0.4156	0.4176	
Step 2-2 GPTQ + clip + AWQ	26.57	29.88	0.3694	0.3700	
Step 3 KL knapsack	26.50	29.72	0.3832	0.3897	worst seed 0.3629
Step 4 W4A8	26.57	29.91	0.3396	0.3430	bonus
Official AutoAWQ	26.57	33.99	0.3692	0.3817	

Requirement: 25 to 33%, MMLU $\geq$ 0.36 (lm_eval [13], limit 0.1, max_len 512). Seed means.

All methods tried: leaderboard

Method	Family	Size %	ppl ↓	MMLU 0.1 ↑	MMLU full ↑	n
S2-1 RTN + clip (MMLU-max alternative)	range search	26.57	31.53	0.4156	0.4176	3
FSQ-B (fixed grid + learned transform) [9]	learned grid	26.57	30.87	0.3928	0.4041	3
AutoRound [5]	learned rounding	26.57	30.16	0.3853	0.3937	3
S3 mixed 26.5 (submitted)	GPTQ+clip+AWQ+knapsack	26.50	29.72	0.3832	0.3897	3
QLoRA r16 [7]	LoRA, adapter counted	28.86	29.14	0.3808	0.3777	2
QA-LoRA r16 [8]	LoRA, folded to grid	26.57	29.20	0.3706	0.3726	2
S2-2 GPTQ + clip + AWQ	learning-free PTQ	26.57	29.88	0.3694	0.3700	4
Official AutoAWQ checkpoint [14]	external anchor	26.57	33.99	0.3692	0.3817	1
KD-QAT + 25% MC-format data	QAT	26.57	28.70	0.3618	0.3631	3
S4 W4A8 per token	W + A	26.57	29.91	0.3396	0.3430	3
S1 RTN	baseline	26.57	34.53	0.3392	0.3414	1
k-means codebook + GPTQ [6]	learned grid	26.26	34.30	0.3366	0.3352	3
KD-QAT, generic text	QAT	26.57	28.54	0.3319	0.3288	3
Hadamard rotation + RTN [4]	rotation (QuaRot R1)	26.57	50.54	0.2874	0.3020	1

RTN + clip beats the submission on MMLU; KD-QAT beats it on ppl. QLoRA adapter counted in size.

Conclusion

- 1. All requirements met.** 26.50% (3.77×), MMLU 0.3832 with every seed ≥ 0.36 ; beats the official AutoAWQ reference on every axis at equal size.
- 2. Clip is a regularizer on outlier weights, and a small model rewards it.** MMLU exceeds FP16, but the ppl gain is small: MMLU and ppl respond to different things.
- 3. GPTQ + AWQ on the clipped grid gives the balanced point.** Replacing RTN rounding brings ppl within 1.2 of FP16 while MMLU stays above the requirement; KL knapsack then improves both.
- 4. Activation quantization is the hard part, not weights.** Dynamic, scale axis limited to per-tensor or per-token, not pre-compensable, and 2642× outlier channels. W4 holds; A8 already costs MMLU.
- 5. At 0.6B, MMLU is a churn metric.** Any perturbation flips 22 to 24% of items; rankings use seed means and gaps under 0.02 are ties. RTN + clip kept as the MMLU-max alternative.

Additional. RTN + clip + QA-LoRA folded

Method

- Base: RTN + clip (W4 g128 asym), weights and grid frozen. MMLU-best but ppl 31.5
- QA-LoRA r16 [8]: input pooled per group of 128, so the merged ΔW is constant within a group. KD to the FP16 teacher, 1000 steps, calibration data only, ~3.5 min
- Folding: the group constant δ is absorbed into the existing zero point, $s(q-z)+\delta = s(q-(z-\delta/s))$. +0 bytes, size stays 26.57%

Base decides the outcome (same QA-LoRA, different base)

Base	ppl	MMLU by seed
S2-2 GPTQ + clip + AWQ	29.20	0.409 / 0.332 (collapse)
AutoRound	29.31	0.417 / 0.379 (unstable)
S2-1 RTN + clip	29.78	0.434 / 0.435 / 0.433

The adapter inherits the base's character: a group-constant correction preserves clip's shrinkage while removing systematic error.

RTN + clip → + QA-LoRA folded (3 seeds)

31.53 → 29.78

ppl (submitted S3: 29.72)

0.4156 → 0.4341

MMLU 0.1, worst seed 0.4329, spread 0.001

0.4365

MMLU full, every seed above FP16 0.4015

References

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